

# Thesauros — AI Layer over PSO

**Audience:** CEO / CTO    **Status:** Product concept    **Scope:** Architecture & governance (not implementation)

## 1. Executive summary

Thesauros is a yield vault that allocates capital across lending protocols using PSO. An AI layer sits above the optimizer: it continuously rebuilds *how* allocation is searched — market regime, risk penalties, and swarm parameters — using the vault's own history and external market signals. PSO still computes percentage weights. Capital movement remains with an operator or Safe multisig.

**Positioning:** AI manages allocation strategy and PSO inputs — not signing keys.

### Differentiation vs a plain APR rebalancer

- PSO input is a risk-adjusted forecast, not a raw on-chain deposit rate
- Optimizer settings are retuned from data and backtests, not hardcoded forever
- Every proposal is explainable and measured against a baseline
- Capital execution is explicitly human- or multisig-gated

## 2. Core idea

| Component  | Role                                                                         |
|------------|------------------------------------------------------------------------------|
| Data       | Vault history (rates, TVL, rebalances, backtests) + external market features |
| AI layer   | Forecast, risk scoring, regime/meta tuning, optional explanation             |
| PSO        | Searches allocation percentages under AI-provided objective and parameters   |
| Governance | Operator / Safe confirms; on-chain vault executes                            |

AI proposes and validates via backtests. AI does not submit transactions.

## 3. Architecture — three layers

### Layer 1 — Forecast + Risk (ML)

Two focused models trained on vault data and market features:

- **Forecast** — expected APR per provider over a short horizon
- **Risk** — instability / drawdown likelihood of the current rate signal

PSO receives a **risk-adjusted signal** aligned to the vault profile (e.g. conservative / pro), not the raw deposit rate.

### Layer 2 — Meta / regime + PSO tuning (ML + offline backtests)

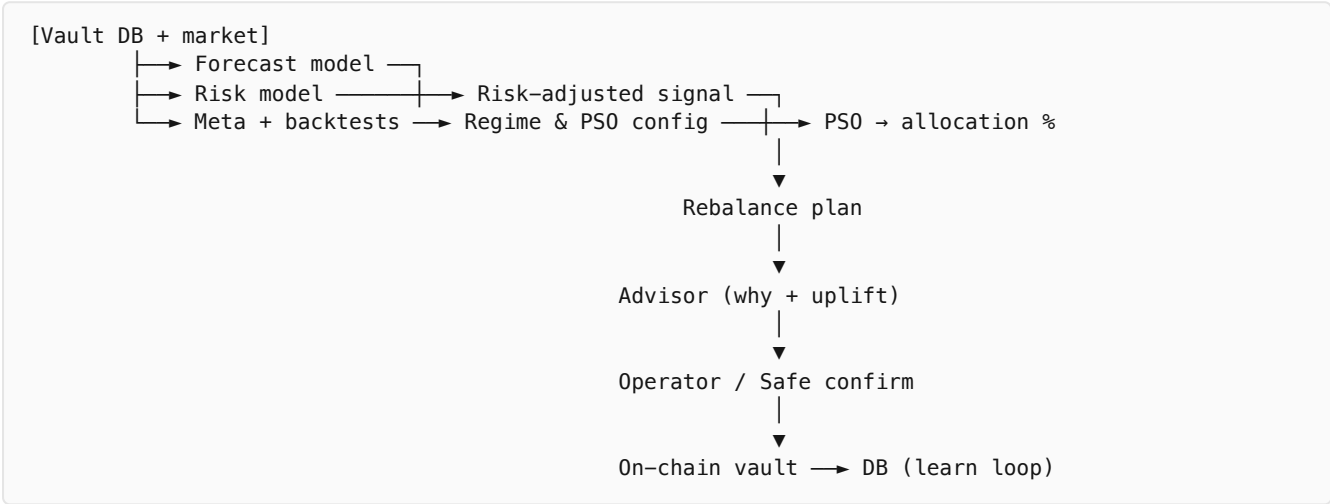
A meta model or scheduled offline loop selects market regime, updates PSO hyperparameters, and adjusts provider weights / penalties. Production PSO remains in place: the AI layer changes configuration

and objective, it does not replace the swarm with a black-box allocator. Feedback loop: new on-chain outcomes → retrain / re-run backtests → updated coefficients.

**Layer 3 — Advisor / explain (optional LLM, non-custodial)**

Formats model outputs and the rebalance plan into a human-readable proposal: what changed, why, expected uplift. Does **not** approve capital moves and does **not** send transactions. Leaves a clear trail: AI proposal vs operator/Safe confirmation.

**4. Operating principle**



- 1. From vault and market data, compute APR forecast and risk per provider.
- 2. Meta loop selects regime and retunes PSO settings via backtests.
- 3. PSO searches weights on the adjusted signal, not raw rate.
- 4. Advisor packages the proposal and rationale.
- 5. Operator/Safe confirms → on-chain execution → outcome returns to the database for the next learning cycle.

**5. Messaging**

|                     |                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------|
| External            | AI manages allocation strategy. Capital execution stays with human control or multisig.    |
| Internal (accurate) | AI manages strategy inputs and PSO configuration. It does not hold or use signing keys.    |
| Proof points        | APR uplift vs baselines (Aave-only / legacy PSO); decision log; explicit human-in-the-loop |

**6. Governance boundary**

| AI does                         | AI does not                    |
|---------------------------------|--------------------------------|
| Forecast and risk scoring       | Sign or broadcast transactions |
| Regime selection and PSO tuning | Final capital decisions        |
| Proposal + explanation          | Bypass RiskGuard / Safe        |

Learn from DB, market, and backtests

Act as autonomous custodian

## 7. Out of scope

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- Model algorithm selection and training pipeline
- Service topology and code changes in the rebalance engine
- Product brand name for the AI layer
- Fully autonomous capital execution

## 8. Short brief

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Thesauros is a yield vault with an AI layer over PSO. Models trained on vault history and market data rebuild the allocation objective and swarm parameters; PSO computes percentages; an operator or Safe moves capital. AI proposes and validates with backtests — it does not sign transactions.

Forecast + Risk → Meta → Advisor

Strategy, not keys

Uplift + decision log + HITL